

The empirical recurrence for OEIS A252286, recovered from its tail

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Abstract

OEIS A252286 records “Empirical recurrence of order 89” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 107$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{89} = 256 \neq 0$, so no further tail satisfies anything shorter; any order-89 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A252286 is “Number of $(n+2) \times (1+2)$ 0..2 arrays with every 3×3 subblock row, column, diagonal and antidiagonal sum not equal to 0 1 or 4.”. It has offset 1 and begins

564, 1130, 2525, 5910, 14042, 33171, 79607, 192757, ...

The entry states:

Empirical recurrence of order 89 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:01:06, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 107$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 107$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 89: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 89.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 214$ exact terms from $n = 3$ on, it returns order 89 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	3	c_{24}	−116159	c_{47}	−1856440	c_{70}	−1898050
c_2	−3	c_{25}	287889	c_{48}	2990852	c_{71}	−835100
c_3	33	c_{26}	−398848	c_{49}	−6257105	c_{72}	−815804
c_4	−81	c_{27}	356774	c_{50}	3805086	c_{73}	−995180
c_5	79	c_{28}	−263360	c_{51}	−7953978	c_{74}	−49248
c_6	−448	c_{29}	33189	c_{52}	5490472	c_{75}	962088
c_7	833	c_{30}	333282	c_{53}	−5735746	c_{76}	603000
c_8	−811	c_{31}	−708931	c_{54}	7074769	c_{77}	114248
c_9	3110	c_{32}	919068	c_{55}	−1373168	c_{78}	100976
c_{10}	−3743	c_{33}	−1371615	c_{56}	5508491	c_{79}	−48536
c_{11}	3885	c_{34}	1053515	c_{57}	2623079	c_{80}	−97696
c_{12}	−10592	c_{35}	−348892	c_{58}	2157348	c_{81}	45040
c_{13}	3591	c_{36}	524355	c_{59}	−1311317	c_{82}	83312
c_{14}	−6618	c_{37}	1050066	c_{60}	−4746831	c_{83}	−23616
c_{15}	8020	c_{38}	−728040	c_{61}	−3109970	c_{84}	−47072
c_{16}	28891	c_{39}	1584497	c_{62}	−1805077	c_{85}	11968
c_{17}	−11689	c_{40}	−1954881	c_{63}	−672936	c_{86}	14720
c_{18}	50885	c_{41}	−182644	c_{64}	−334697	c_{87}	1408
c_{19}	−94112	c_{42}	−1231468	c_{65}	562897	c_{88}	384
c_{20}	40047	c_{43}	−845383	c_{66}	4079900	c_{89}	256
c_{21}	−103801	c_{44}	1082647	c_{67}	2687507		
c_{22}	23702	c_{45}	−729403	c_{68}	705500		
c_{23}	97472	c_{46}	3786262	c_{69}	−1816306		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A252286 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{89} - c_1x^{89-1} - \dots - c_{89}$. Its constant term is $-c_{89} = -256$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 89 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 89, so $\deg q = 89$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 89, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 89, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 256.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-89 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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